Nokia Core Node Maintenance Procedure for Optical Transport Failure on BLR-CORE-02

Node: BLR-CORE-02 | **Vendor:** Nokia | **Severity:** Warning | **Alarm Code:** ALARM_2808

Description

The Nokia Core Node, specifically the BLR-CORE-02 instance operating at an unknown site, may experience Optical Transport Failure. This type of failure is primarily attributed to the degradation of SFP/XFP components or physical damage to the fiber optic cables, leading to a reduction in receive (Rx) and transmit (Tx) power levels. The ALARM_2808, categorized as a Warning severity, is triggered when the system detects a deviation from the predetermined power thresholds. This SOP is designed to guide the technician through a series of diagnostic and resolution steps to identify and rectify the root cause of the failure, ensuring the continued operation of the Nokia Core Node within the constraints of its optical transport capabilities.

Pre-requisite Checks

- check: Verify the site location and accessibility for remote hands or on-site personnel.

- check: Confirm the status of the redundant link to ensure it is operational and not contributing to the failure.

- check: Gather and review the historical alarm logs to identify potential recurring issues.

Potential Root Causes

Cause	Probability
Degraded SFP/XFP components	High
Physical damage to fiber optic cables	Medium
Rx/Tx power level misconfiguration	Low

Diagnostic Commands & Expected Outputs

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Command: show controllers optics, Expected Output: Optics information for the affected interface, Description: This command displays the optics-related information, including the SFP/XFP details, which can help identify potential degradation or faults.

Command: show interface , Expected Output: Interface information, including Rx/Tx power levels, Description: This command provides an overview of the interface's operational status, including the current Rx and Tx power levels, which can help determine if the issue is related to a power misconfiguration.

Command: show mpls ldp , Expected Output: LDP information for the affected interface, Description: This command displays the LDP-related information, which can help identify potential issues with LDP
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signaling, contributing to the optical transport failure.
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Resolution Steps

Step 1: Step: Replace the degraded SFP/XFP, Command: replace module , Description: Physically replace the SFP/XFP with a new one, ensuring the correct part number and compatibility with the Nokia Core Node.

Step 2: Step: Verify and adjust Rx/Tx power levels, Command: set interface rx-tx-power-adjust , Description: Verify the current Rx and Tx power levels and adjust them as necessary to ensure optimal operation and prevent further failure.

Step 3: Step: Inspect and repair the fiber optic cables, Command: inspect fiber-optic-cables, Description: Perform a visual inspection of the fiber optic cables and repair or replace them as necessary to restore the connection and prevent further failure.

Post-Check Verification

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Command: show controllers optics , Expected Output: Optics information confirming the resolution, Description: Verify that the SFP/XFP replacement or adjustment has resolved the optical transport failure, and the optics information confirms the expected values.

Command: show interface , Expected Output: Interface information confirming the resolution, Description: Verify that the Rx and Tx power levels are within the acceptable range, and any adjustments made have resolved the issue.

Command: show mpls ldp , Expected Output: LDP information confirming the resolution, Description: Verify that the LDP signaling is stable and functioning as expected, contributing to the resolution of the optical transport failure.
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Roll-back Procedures

- Step: Restore the original SFP/XFP, Command: replace module revert, Description: Revert the SFP/XFP replacement to its original state, ensuring the correct part number and compatibility with the Nokia Core Node.

- Step: Revert Rx/Tx power levels, Command: set interface rx-tx-power-adjust , Description: Revert the Rx and Tx power levels to their original values, ensuring optimal operation and preventing further failure.

- Step: Reinspect and repair the fiber optic cables, Command: inspect fiber-optic-cables revert, Description: Reperform the visual inspection of the fiber optic cables and repair or replace them as necessary to restore the connection and prevent further failure.